RESEARCH TALK

Can Foundation Models Really Understand Economic Research?

A Pipeline-Level Benchmark for Agentic Research Systems

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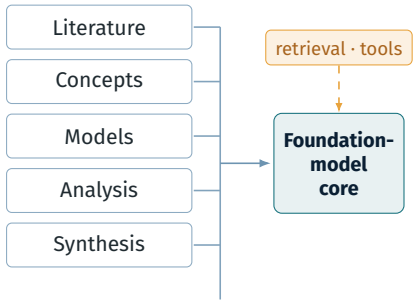
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Why a pipeline-level benchmark?

Economic research is a chain of intellectual stages — literature positioning, concept grounding, formal modeling, analysis, and synthesis. Agentic systems increasingly automate this chain, yet existing benchmarks mostly test isolated skills or non-academic scenarios.

The missing diagnostic

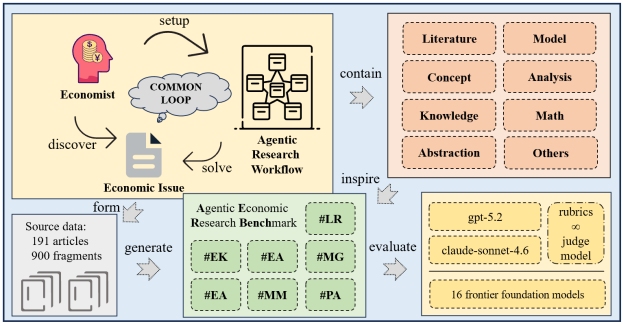
Can a foundation-model core carry each research stage on its own — and where must an agent add retrieval, tools, or human oversight?



An agent orchestrates around the model core.

OVERVIEW

AER-bench at a glance



An example in AER-bench

Item: MM_W2341847281_Q1
Type: MM - Equation design
Difficulty: Medium
Question: An analyst studies TV-station valuation, where station-level cash flow follows a structural equation [given], with an additive Gaussian shock parameterized by θ. True cash flow is unobserved at the station level; only group-year empirical moments [given]—mean, three quartiles, and mean non-broadcast revenue—and their simulated counterparts are available. Task: write a simulated minimum-distance objective $Q(\theta)$ matching all three sets of moments, and define the estimator $\hat{\theta}$

Rubrics:

Dimension	Criterion
Formula correctness	Equivalent to reference; derivations valid.
Notation consistency	Symbols match the problem setup throughout.
Derivation completeness	Intermediate steps shown and verifiable.
Economic intuition	Interprets the math in economic terms.

191
top-journal articles

900
historical text segments

1,629
evaluation items

16
frontier models

Seven measurable modules reverse-engineer the intellectual core of economic research.

TASKS

Seven tasks cover the research pipeline

Code	Task	What it tests
EK	**Economic Knowledge**	paper-specific methods and findings
LR	**Literature Review**	argument structure and citations
CI	**Concept Identification**	eight economic properties of a passage
MG	**Model Generation**	objectives, constraints, equilibrium
MM	**Mathematical Modeling**	derivation and economic intuition
EA	**Economic Analysis**	mechanism, theory, policy
PA	**Paper Abstraction**	accurate academic synthesis

Benchmark boundary search and retrieval disabled · no code execution or file handling · analytical core only

The goal is a module-level diagnostic for building research agents.

DATA

Data construction: frontier + historical memory

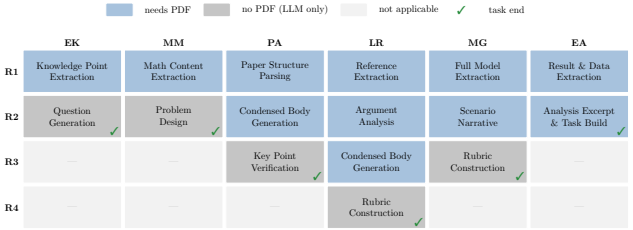
191 modern articles

Open-access 2025 papers from the *American Economic Review* and *Econometrica*.

900 historical segments

1900–1950 texts, manually annotated by economics PhDs.

Six modern tasks use multi-round PDF extraction; CI uses expert labels over eight dimensions.



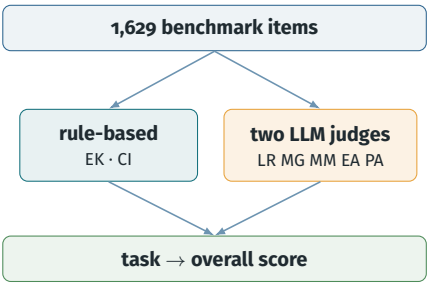
Generation depth: 2 rounds (EK, MM, EA) up to 4 rounds (LR).

1,629 items across seven tasks scored module by module, then aggregated with equal category weights.

Evaluation protocol: objective where possible

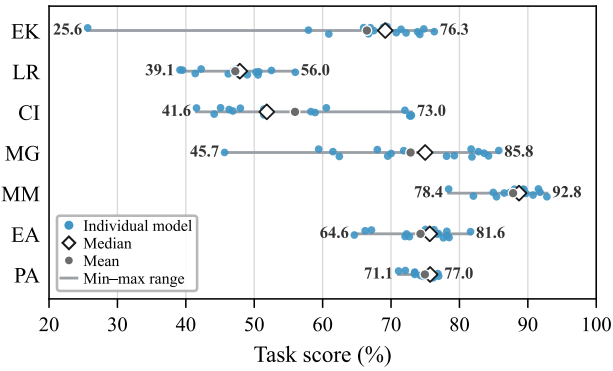
Hybrid scoring

- **Rule-based:** EK and CI use exact or normalized matching.
- **LLM-as-judge:** LR, MG, MM, EA, and PA use dimension-level rubrics.
- Two judges (Claude 4.6 Sonnet, GPT-5.2) are averaged; agreement is $R^2 = 0.97$.
- All seven task categories receive equal overall weight.



zero-shot · 16 frontier models · tool-free baseline

Overall results: formal strength, uneven knowledge



Task-level score distribution across 16 models.

OVERALL SCORE

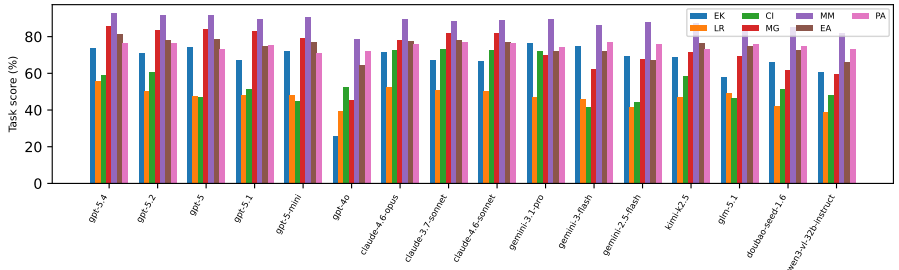
GPT-5.4	**75.0%**
Claude 4.6 Opus	**74.0%**

Statistically indistinguishable ($p = 0.099$).

- **MM** is a relative strength, with a median near 90%.
- **LR** is the bottleneck: the best model reaches 56.0%.
- **EK** spans 25.6–76.3% across models.

RESULTS

Capability fingerprints differ by model family



Per-task scores reveal profiles that an overall average hides.

OpenAI formal strength

GPT-5 variants hold the top five MM scores (89.7–92.8%).

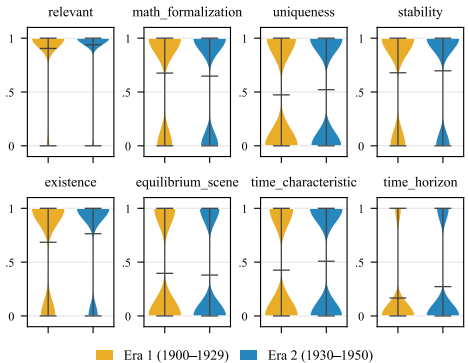
Anthropic concept edge

Claude clusters at 72.7–73.0% on CI, over 12 pts above GPT-5.2.

Google uneven profile

Gemini is competitive on EK and PA but trails on LR, MG, and EA.

Historical concepts expose an uneven blind spot



CI across eight concept dimensions, 1900–1929 vs. 1930–1950.

What the data say

Aggregate performance is about **5 pp lower** on earlier text, led by `equilibrium_scene` and `existence`.

- The effect is model-stratified, not universal.
- Controlling for chunk length: $+2.75$ pp, $p = 0.442$.
- The descriptive gap concentrates in the longest quartiles: $+9.2$ and $+12.7$ pp.

IMPLICATIONS

What AER-bench changes for agentic research

01

Augment knowledge

Retrieval and citation databases matter most for EK and LR, where parametric knowledge is volatile.

02

Route by capability

Family fingerprints support module-aware model routing, not one-model-fits-all pipelines.

03

Test historical reach

CI diagnoses representational gaps in the history of economic thought.

Take-home message

Models have largely mastered formal manipulation and synthesis, while knowledge-dependent stages still need augmentation. AER-bench supplies the tool-free reference layer against which complete agentic systems can be measured.

Thank you **Questions?**

Full task inventory and item shares

Task	Items	Share	Scoring	Core capability
EK	189	11.6%	Rule-based	domain knowledge
LR	187	11.5%	Judge + citation recall	literature positioning
CI	160	9.8%	Rule-based	conceptual grounding
MG	124	7.6%	LLM judge	model construction
MM	399	24.5%	LLM judge	symbolic derivation
EA	380	23.3%	LLM judge	mechanism and policy
PA	190	11.7%	LLM judge	academic synthesis
Total	**1,629**	**100%**		

Dimension weighting

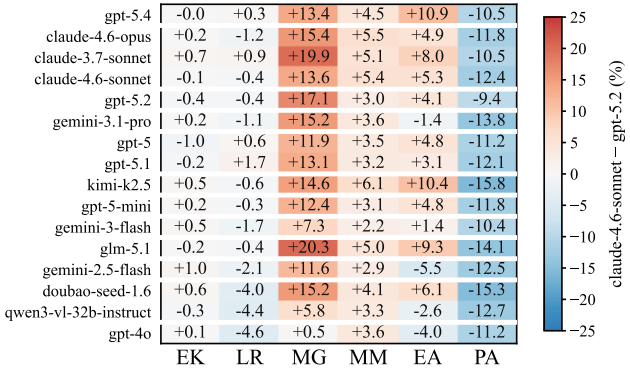
All seven categories receive equal weight; intra-task dimensions use predefined rubric weights (paper Table 2).

Complete overall ranking

Family	Model	Overall [95% CI]	EK	MM	PA	LR	MG	EA	CI
OpenAI	gpt-5.4	**75.0** [74.0,76.1]	73.9	92.8	76.2	56.0	85.8	81.6	58.9
OpenAI	gpt-5.2	73.0 [72.1,74.0]	70.8	91.5	76.4	50.3	83.6	78.2	60.5
OpenAI	gpt-5	71.0 [69.9,72.0]	74.2	91.8	73.5	47.7	84.2	78.5	46.9
OpenAI	gpt-5.1	70.0 [69.0,71.0]	67.5	89.7	75.6	48.1	82.8	75.0	51.4
OpenAI	gpt-5-mini	69.0 [68.0,70.0]	72.2	90.8	71.1	48.1	79.2	76.8	45.1
OpenAI	gpt-4o	54.0 [53.1,55.0]	25.6	78.4	72.1	39.5	45.7	64.6	52.3
Anthropic	claude-4.6-opus	74.0 [73.0,74.9]	71.5	89.4	76.1	52.5	78.1	77.7	72.7
Anthropic	claude-3.7-sonnet	73.7 [72.8,74.6]	67.0	88.6	76.8	50.7	81.7	78.2	73.0
Anthropic	claude-4.6-sonnet	73.4 [72.5,74.3]	66.7	88.8	76.3	50.6	81.8	76.9	72.9
Google	gemini-3.1-pro	71.6 [70.6,72.5]	76.3	89.3	74.3	47.0	70.0	72.0	72.0
Google	gemini-3-flash	65.8 [64.9,66.8]	74.8	86.5	77.0	46.2	62.4	72.3	41.6
Google	gemini-2.5-flash	64.9 [63.9,65.9]	69.4	88.0	76.1	41.4	68.0	67.1	44.1
Others	kimi-k2.5	69.1 [68.0,70.1]	68.9	87.4	73.5	47.3	71.9	76.3	58.3
Others	glm-5.1	65.5 [64.5,66.6]	57.9	85.4	75.8	49.0	69.5	74.8	46.3
Others	doubao-seed-1.6	64.8 [63.8,65.8]	66.0	84.9	74.7	42.2	61.5	72.7	51.4
Others	qwen3-vl-32b	61.3 [60.3,62.3]	60.9	82.0	73.5	39.1	59.4	66.2	48.0

Full task columns and bootstrap intervals are reported in the paper appendix.

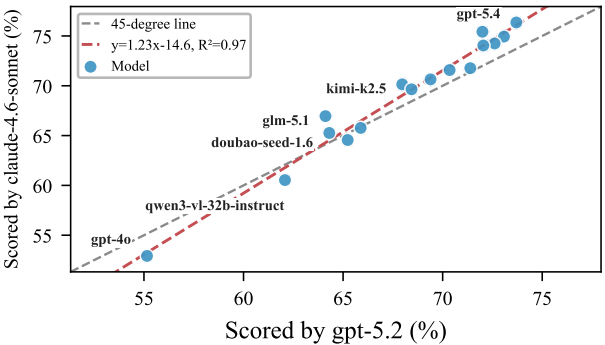
Judge differences and robustness



Claude 4.6 Sonnet score minus GPT-5.2 score, by task and model.

- Judge offsets vary by task: Claude scores some MG outputs higher, while GPT scores PA higher.
- Family-level direction: Anthropic $+2.90$ pp vs. OpenAI $+1.41$ pp.
- Mann–Whitney $p = 0.095$: evidence does not reach the 0.05 threshold.
- Averaging two judges attenuates, but does not eliminate, evaluator concerns.

Cross-judge agreement



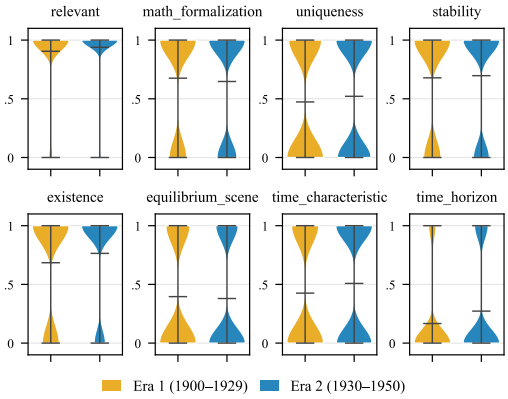
Overall scores from Claude 4.6 Sonnet against GPT-5.2 across 16 models.

Agreement

Linear fit: $R^2 = 0.97$; mean absolute difference 1.6 pp.

- The two judges track closely across all 16 models.
- Dimension-level rubrics and averaging improve robustness.
- Task-level offsets still warrant inspection.

Historical robustness, limits, and next steps



Concept-identification scores by era and concept dimension.

- Chunk-length control reduces the era coefficient to $+2.75$ pp ($p = 0.442$); longer chunks are harder.
- The era gap concentrates in the longest quartiles, and model patterns are heterogeneous.
- **Limits:** extraction cost, PDF notation artifacts, predominantly English corpus, no human baseline, residual judge bias.
- **Next:** test retrieval and code augmentation, add an independent third judge, and run length-controlled historical comparisons.